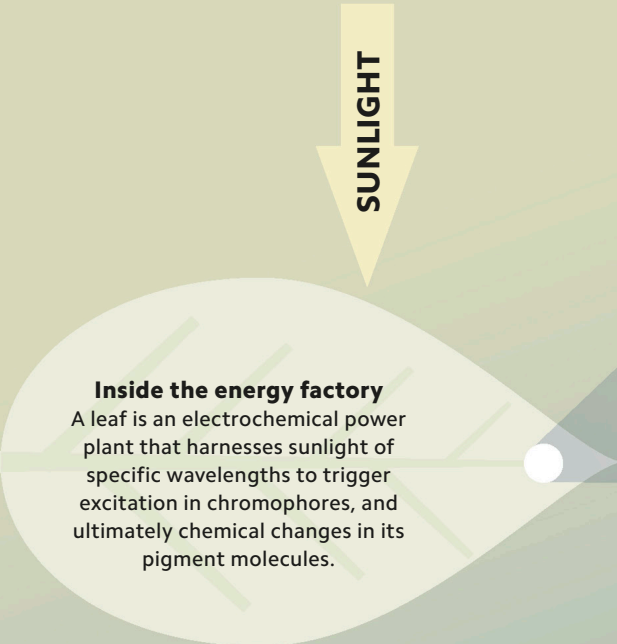


PHOTONS AND FOLIAGE

Photosynthesis is the process by which plants manufacture sugars and other chemicals using energy from sunlight. Its first step involves photons triggering chemical changes to molecules called chromophores. Energy produced by these changes is transferred to other molecules, where it can be put to use with remarkable efficiency, involving synchronized vibration between different energy states. Many biologists believe that photosynthesis has evolved to take advantage of the quantized nature of light energy, and some go further, suggesting that other quantum phenomena, such as superposition (see pp.38–39), may play a role.



SUNLIGHT

Inside the energy factory

A leaf is an electrochemical power plant that harnesses sunlight of specific wavelengths to trigger excitation in chromophores, and ultimately chemical changes in its pigment molecules.